day 28 prep

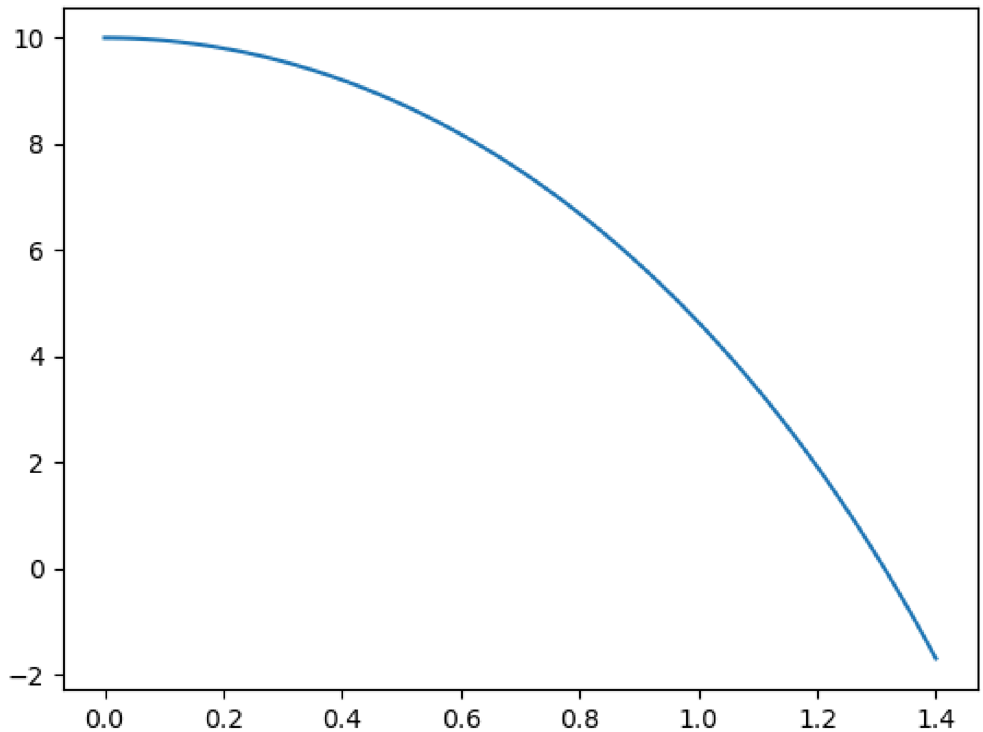
Ok I fixed a problem from day 26 that was nagging me... this v^2 drag dependence was a problem and that is because of the sign of the velocity.

In [1]:
```
%matplotlib ipympl
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
# define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    v = r[1]
    g = 9.8 # N/kg
    m = 1 # kg
    c = 0.05 # what are these units?
    fx = v  # this is the dx/dt equation!
    fv = -g - c/m*(v**2) # this is the dv/dt equation!
    return(np.array([fx, fv]))

# define the boundary condition
a = 0.0
b = 1.414
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([10.0, 0.0]) # initial condition

tpoints = np.arange(a, b, dt)

xpointsD2 = []
vpointsD2 = []

for i in tpoints:
    xpointsD2.append(r[0])
    vpointsD2.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig0, ax0 = plt.subplots()
ax0.plot(tpoints, xpointsD2)
#ax0.plot(tpoints, xpointsD)
#ax0.plot(tpoints, xpoints)
```

Out[2]:
```
[<matplotlib.lines.Line2D at 0x7ae222ce0cb0>]
```

Figure




```
In [ ]:
```